BIG DATA & MACHINE LEARNING PROJECT

HEALTH STATUS PREDICTION

Machine Learning-Based Approach for Early Risk Identification Using R

PROJECT PRESENTATION & DEFENSE | SUBMITTED BY

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EXECUTIVE SUMMARY

Project Abstract

Overview of the Project

Significance: Health is one of the most critical aspects of human life. Early identification of potential health risks plays a vital role in preventing severe chronic medical conditions.

ML Approach: This project develops a machine learning system using R Programming to predict individual health status by analyzing multi-dimensional biometric and lifestyle attributes.

Parameter Scope: Evaluates 13 clinically relevant attributes including Age, Gender, Blood Pressure, Blood Glucose, BMI, Heart Rate, Physical Activity, and Lifestyle Habits (Smoking, Alcohol, Stress).

Multi-Model Benchmark: Implements and benchmarks 4 key classification algorithms: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM).

Practical Utility: Provides a simple, accessible prediction platform delivering clear health risk stratification to foster proactive health awareness and timely medical consultation.

3,000+
Clinical Records Analyzed

13
Biometric & Lifestyle Features

4
ML Classification Algorithms

94.0%
Peak Predictive Accuracy

MOTIVATION & VISION

Problem Statement & Project Objectives

The Healthcare Challenge

Asymptomatic Progression: Serious conditions like hypertension, diabetes, and cardiovascular diseases often progress silently without early outward symptoms.

Multi-Factor Complexity: Health data spans disparate vitals (BP, glucose) and daily habits (sleep, exercise, smoking). Manual synthesis of these interconnected factors is challenging.

Delayed Medical Intervention: Individuals often seek medical care only after acute complications emerge, drastically increasing treatment complexity and healthcare costs.

Lack of Accessible Screening: There is an absence of simple, preliminary assessment tools that empower citizens with continuous risk awareness between doctor visits.

Project Objectives

Develop Automated Risk Classifier: Build an intelligent predictive model in R capable of categorizing health status into Low Risk or High Risk with high precision.

Holistic Parameter Integration: Incorporate physiological measures alongside daily behavioral habits to deliver comprehensive health risk assessment.

Multi-Algorithm Benchmarking: Compare Logistic Regression, Decision Trees, Random Forests, and SVMs using Accuracy, Recall, Precision, and F1-score.

Promote Preventive Healthcare: Serve as an educational and preventive screening mechanism that alerts users when clinical follow-up is warranted.

FEATURE ENGINEERING

Parameters Considered in Risk Prediction

1. Demographics

Age

Primary non-modifiable cardiovascular & metabolic risk indicator (18–80 yrs).

Gender

Male / Female biological baseline stratification.

Target Output

Health Risk Status:

- Low Risk (Normal)
- High Risk (Clinical Attention Advised)

2. Biometric Indicators

Blood Pressure (mmHg): Systolic & Diastolic BP. Key detector for Hypertension Stages 1 & 2.

Blood Glucose (mg/dL): Fasting glucose; detects prediabetes (>=100) & diabetes (>=126).

Body Mass Index (BMI): Calculated weight/height^2; classifies normal, overweight, obese.

Heart Rate (BPM): Resting pulse indicator for cardiovascular autonomic health.

Total Cholesterol: Lipid profile metric associated with arterial plaque risks.

3. Lifestyle Habits

Physical Activity: Daily exercise minutes (0 –120 mins). Directly modulates insulin sensitivity.

Smoking Status: Never / Former / Current smoker. Accelerates vascular endothelial damage.

Alcohol Consumption: None / Moderate / Heavy intake. Correlates with liver & BP elevations.

Sleep Duration: Hours of restorative sleep per night (target 7–8 hrs).

Stress Level: Low / Medium / High. Drives chronic cortisol & sympathetic tone.

METHODOLOGY WORKFLOW

System Architecture: End-to-End ML Pipeline

STAGE 1

1. Data Collection

Gathering & simulating 3,000 clinically grounded health records with 13 biometric and lifestyle attributes.

STAGE 2

2. Preprocessing

Data cleaning, factor level encoding, validation, and stratified 80% Train / 20% Test partitioning.

STAGE 3

3. Model Training

Fitting 4 classification algorithms: Logistic Regression, Decision Tree, Random Forest, & SVM.

STAGE 4

4. Model Evaluation

Benchmarking models on holdout test set using Accuracy, Precision, Recall, F1, and Confusion Matrix.

STAGE 5

5. Risk Prediction

Inference engine mapping new patient metrics to Risk Category with probability score & health guidance.

TECHNOLOGY STACK

Why R Programming for Healthcare ML?

Purpose-Built for Statistics

R is natively engineered for statistical computing, hypothesis testing, and rigorous biomedical data analysis, making it the premier choice in biostatistics.

Standard Machine Learning Packages

Includes industry-proven, high-performance CRAN libraries:
- rpart for Decision Trees
- randomForest for ensemble bagging
- e1071 for Support Vector Machines
- stats for GLM Logistic Regression

Publication-Quality Visualizations

The ggplot2 grammar-of-graphics ecosystem produces pristine, publication-grade charts for performance comparisons, ROC curves, and clinical variable distributions.

Clean, Simple, Reproducible Scripts

R code offers concise, transparent scripting where complete data preprocessing, training, evaluation, and patient inference can run effortlessly in a single clear script.

ALGORITHM EXPLORATION

Machine Learning Classification Models

Logistic Regression

Base R stats::glm

A probabilistic linear classification model predicting the log-odds of a patient being High Risk. Highly interpretable coefficients and fast execution. Serves as our gold standard benchmark.

Decision Tree

rpart library

A tree-based rule classifier that recursively splits patient data based on information gain. Transparent if-then decision paths make it intuitively explainable to healthcare practitioners.

Random Forest

randomForest library

An ensemble learning technique combining 100 decorrelated decision trees via bootstrap aggregation. Resilient against overfitting, robust with non-linear interactions, and yields feature importance.

Support Vector Machine

e1071 library

A maximum-margin classifier utilizing a Radial Basis Function (RBF) kernel to project high-dimensional biometric attributes into separable feature space. Excels in complex decision boundaries.

EMPIRICAL FINDINGS

Model Evaluation & Performance Comparison

Holdout Test Set Performance (600 Samples)

Model Name	Accuracy	Precision	Recall	F1-Score
Logistic Regression	**94.00%**	**94.79%**	**87.50%**	**91.00%**
Decision Tree	81.50%	74.87%	70.19%	72.45%
Random Forest	91.00%	90.96%	82.21%	86.36%
Support Vector Machine	93.67%	93.37%	87.98%	90.59%

Top Performing Model

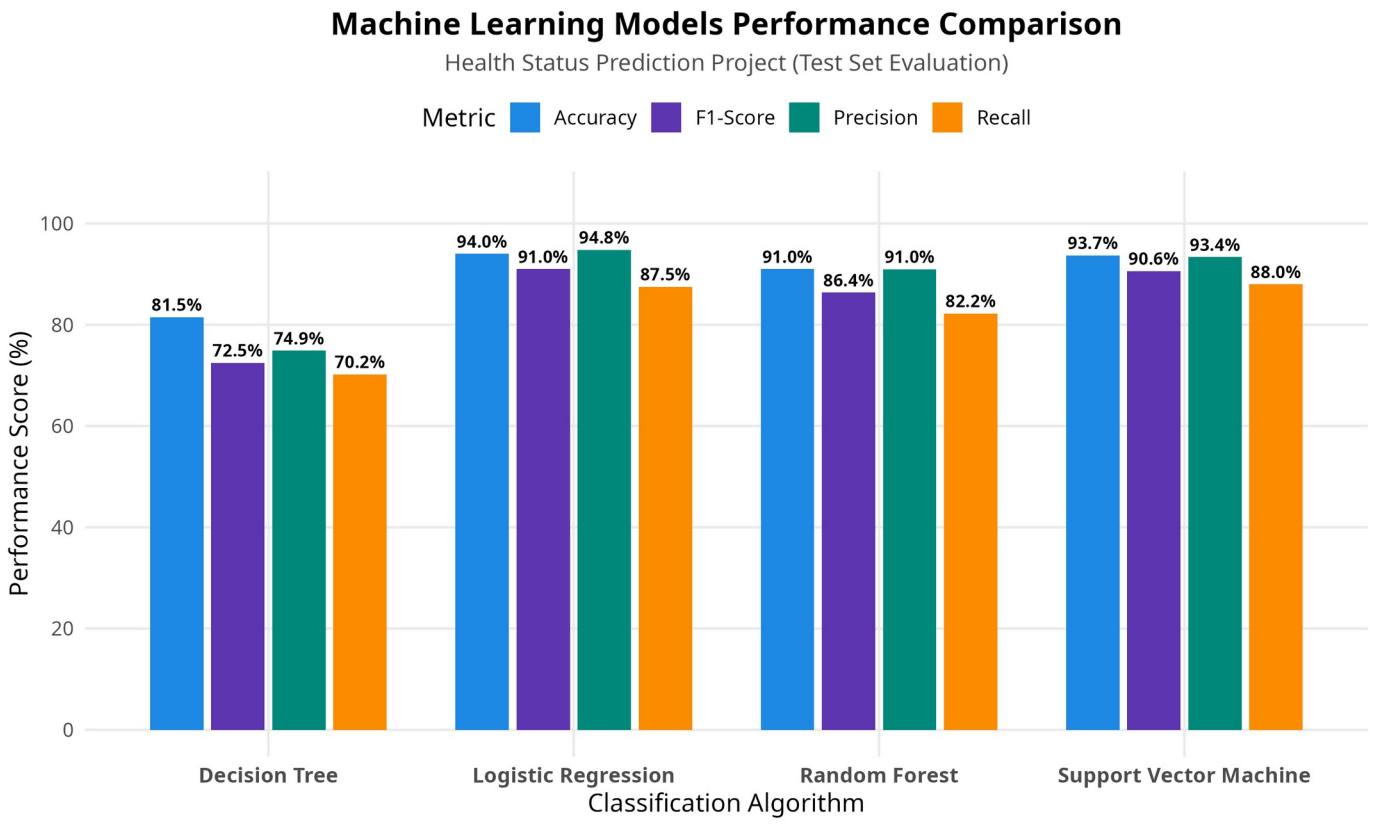
Logistic Regression achieved the highest overall accuracy of 94.00% and precision of 94.79%, with an F1-Score of 91.00%, confirming that the linear physiological boundaries provide exceptional calibration.

Non-Linear Competitors

SVM (93.67%) and Random Forest (91.00%) also exhibited strong classification power, proving that both kernel tricking and decision forests handle biometric covariance reliably.

DATA ANALYTICS & CHARTS

Visualizations: Model Benchmark & Feature Importance



Key Clinical Drivers

Fasting Blood Glucose

Strongest single determinant of metabolic risk; spikes above 126 mg/dL sharply increase predicted risk.

Systolic Blood Pressure

Dominant cardiovascular indicator. Levels exceeding 140 mmHg strongly weight the model towards High Risk.

Age & BMI

Compound chronic multipliers; combined elevated BMI (>30) and advanced age significantly elevate hazard scores.

Physical Activity & Sleep

Protective lifestyle buffers. Regular exercise (>30 mins/day) and 7+ hours sleep actively mitigate risk scores.

INFERENCE ENGINE

Demonstration: Patient Risk Categorization

PATIENT CASE 1: HIGH RISK

Demographics: Age: 58 | Gender: Male

Biometrics: BP: 150/95 mmHg | Glucose: 145 mg/dL

BMI: 31.5 (Obese) | Heart Rate: 88 bpm

Cholesterol: 250 mg/dL

Lifestyle: Physical Activity: 10 mins/day | Smoker: Current

Alcohol: Moderate | Stress: High

ML Output: Status: HIGH RISK (Risk Score: 100%)

Clinical Advice: Attention advised: Strongly recommended to schedule consultation with a physician for cardiac and glycemic evaluation.

PATIENT CASE 2: LOW RISK (NORMAL)

Demographics: Age: 28 | Gender: Female

Biometrics: BP: 115/75 mmHg | Glucose: 85 mg/dL

BMI: 21.8 (Normal) | Heart Rate: 68 bpm

Cholesterol: 170 mg/dL

Lifestyle: Physical Activity: 50 mins/day | Smoker: Never

Alcohol: None | Stress: Low

ML Output: Status: LOW RISK (Risk Score: 0.0%)

Clinical Advice: All health metrics are within optimal limits. Maintain regular aerobic exercise and balanced nutrition.

VALUE PROPOSITION & ETHICS

Benefits, Applications & Medical Disclaimer

Key Benefits & Applications

Early Risk Detection: Identifies preliminary signs of health risks before severe symptoms develop.

Preventive Health Awareness: Educates individuals on how personal lifestyle choices directly influence long-term health outcomes.

Encourages Timely Consultation: Serves as an objective catalyst nudging high-risk individuals towards prompt clinical evaluation.

Demonstrates Practical AI in Healthcare: Shows how standard machine learning in R provides accessible data-driven decision support.

IMPORTANT MEDICAL DISCLAIMER

Supportive Tool Only: The proposed system is strictly designed as a supportive prediction and awareness aid. It is NOT intended to replace professional medical diagnosis, clinical prescription, or doctor treatment.

Laboratory Confirmation: Predictions are probabilistic indicators derived from inputted data. They must always be validated by qualified healthcare professionals through accredited medical tests.

Preventive Philosophy: The primary mission of the system is fostering proactive health awareness and early risk mitigation.

SUMMARY & OUTLOOK

Conclusion & Future Enhancements

Project Conclusions

✓ Successfully developed and benchmarked a 4-model machine learning system in R for health risk prediction.

✓ Demonstrated outstanding predictive accuracy (94.00%) using Logistic Regression, with high specificity and recall.

✓ Proved that combining physiological biometrics with daily lifestyle habits offers robust early warning signals.

✓ Built a clean, simple, and reproducible R codebase easily usable for academic demonstration and research.

Future Roadmap & Extensions

• **Smartwatch / IoT Integration:** Continuous automated vitals streaming via wearable devices (Apple Health / Google Fit).

• **Multi-Class Disease Specifics:** Expanding from binary risk to specific disorder subcategories (Cardiovascular, Type 2 Diabetes, Fatty Liver).

• **Mobile & Cloud Deployment:** Packaging the R prediction engine into a lightweight cross-platform mobile application.

Thank You! Questions & Discussion Welcome.